Appendix C

A Career Map: Roles, Chapters, and Preparation

This book's stated audience is a reader with quantitative training who wants a career in the financial industry, and the chapters have been building the technical vocabulary that industry runs on. What the chapters do not do, because no single chapter is the right place for it, is describe the industry's job landscape: which roles exist, what each one does all day, and which parts of this book matter most for each. This appendix closes that gap. It is a map, not a syllabus; job titles vary considerably across firms (the same work may be called "quantitative researcher" at a hedge fund, "strats" at an investment bank, and "data scientist" at a fintech), so the descriptions below are organized around what the work is rather than what any particular firm calls it.

One framing helps before the table. Broadly, quantitative roles in finance divide into those that *make* risk (research and trading roles, judged on returns), those that *measure and police* risk (risk management, model validation, and compliance roles, judged on rigor and independence), and those that *build the machinery* both sides run on (engineering and data roles, judged on reliability and scale). All three groups hire people with this book's intended background, the mix of statistics, programming, and financial context, but they reward different temperaments, and the interview processes differ accordingly: the first group leans hardest on probability, brainteasers, and market intuition; the second on statistical rigor, regulatory awareness, and the ability to explain a model's limits; the third on software engineering and system design, with the finance tested more lightly.

C.1 Reading the Map

A few patterns in Table C.1 deserve comment. Chapters 2 and 6, Python and machine learning fundamentals, appear implicitly in every row; they are the price of admission everywhere and are omitted from rows only where a role's distinctive preparation lies elsewhere. Chapter 18, explainability and fairness, appears in every role that touches a regulated decision about a person, which is the pattern the chapter itself predicts: the closer a model sits to a consumer outcome, the more its explanation apparatus matters relative to its accuracy. And Chapter 21's capstone appears wherever the answer to "what should I build to demonstrate competence?" is a complete, end-to-end project rather than familiarity with a single method; a worked capstone, cleaned data, calibration checks, and fairness audit included, is the most credible artifact a candidate without industry experience can bring to an interview, because its nine-stage template mirrors how the work is structured inside a firm.

The risk-and-validation column of the industry deserves a special word, because candidates with strong ML backgrounds systematically underrate it. Model validation and risk roles are often easier to enter than front-office research, teach the firm's full model landscape faster than any other seat, are hired against a regulatory requirement rather than a profit forecast (making them notably more stable across market cycles), and increasingly involve hard technical work: challenging a deep-learning surveillance model requires understanding it at least as well as its builder did. A reader whose favorite chapters were 8 through 11 and 18 should take that as information about where they would thrive.

Wulin.Suo@Queensu.ca

Role	What the job actually is	Core chapters
Quantitative researcher (buy side)	Finding and testing predictive signals in market, fundamental, text, and alternative data; building and validating backtests	2, 5–7, 12, 13, 16, 17
Execution and e-trading quant	Designing the algorithms that execute orders and make markets: impact models, order routing, market-making logic	3, 13, 14, 19
Market risk quant	Measuring how much the firm can lose to market moves: VaR and ES models, stress testing, backtesting, limits	5, 7, 8
Credit risk modeler	Estimating who will default and what it costs: PD/LGD/EAD models, scorecards, portfolio credit models, capital	4, 6, 9, 18, 21
Actuary and insurance data scientist	Pricing, reserving, claims, and catastrophe modeling for insurers: GLMs and their machine-learning challengers inside a filed-rating regime	5, 6, 10, 11, 18
Model validator (second line)	Independently challenging every model above: testing assumptions, benchmarking, documentation, SR 11-7/SR 26-2 discipline	6–11, 18
Operational-risk and surveillance analyst	Quantifying and detecting operational failures: loss models, anomaly detection, conduct and communications surveillance	6, 10, 15, 16
Fraud and financial-crime data scientist	Catching fraud, money laundering, and market abuse in transaction and network data, at extreme class imbalance	6, 14, 15, 17, 18
NLP and alternative-data researcher	Turning filings, calls, news, satellite and web data into investable signals with point-in-time discipline	13, 16, 17
Fintech and lending ML engineer	Building production credit, robo-advisory, and payments models end to end, with fairness and explainability constraints	2, 6, 9, 12, 18, 21
AI governance and compliance specialist	Setting and enforcing the rules AI systems must satisfy: model inventories, monitoring, fairness audits, regulatory response	10, 15, 16, 18

Table C.1: A map from common quantitative-finance roles to the chapters that prepare for them

Wulin.Suo@Queensu.ca

C.1.1 Roles That Have Taken Shape by 2026

The ten rows above describe seats that have existed for a decade or more. Table C.2 adds seven that have grown, in the last few years, from a side responsibility into a job title of their own, each mapped to the chapters that prepare for it in the same way.

Each of the seven follows from a development this book has traced. The validator and governance rows exist because the models changed faster than the control functions around them: a validator trained on regressions must now challenge a gradient-boosted credit model, a language model's free-text output, or a policy learned by trial and error, and Chapter 20's discussion of algorithmic collusion and herding shows why an institution running many AI agents needs someone watching what they do together, not only one at a time. The LLM engineer is the production counterpart of Chapter 16's generative-AI sections, where most of the work turns out to be evaluation and control rather than the model itself. Payments fraud became its own specialty as instant-payment systems spread, since a payment that settles in seconds cannot be recalled once a scam is noticed, so the decision has to be right before authorization. The treasury and ALM quant rose in standing after the 2023 failure of Silicon Valley Bank, which Chapter 8 examines as an interest-rate-risk and deposit-run event rather than a credit one. And climate-risk and central-bank research roles have pulled in the scenario analysis, stress testing, and agent-based methods of Chapters 8, 11, and 20. What the seven share is the pattern of the original table: each pairs a specific financial problem with the machine-learning tools now applied to it, and each rewards a candidate who can explain where those tools fail.

C.2 Preparing, Concretely

Three uses of this book map onto how candidates are evaluated. First, the worked examples are interview questions: computing a two-asset portfolio variance by hand, walking through a limit-order-book fill, explaining why the 99.9% operational-loss quantile dwarfs its mean, deriving a Shapley decomposition for a small linear model, and reasoning through Bayes' theorem at a realistic base rate are all standard fare, and every one of them is a worked example somewhere in Chapters 1 through 21. Second, the challenge exercises are portfolio projects: each one pairs a method from a chapter with a real, freely downloadable dataset, which is the scope of take-home assignments many firms now use in place of a second interview round. Third, the Further Reading lists are the canon: Hull for derivatives and risk, Bodie-Kane-Marcus for investments, Chan and Hilpisch for systematic trading, and the original papers behind XGBoost, SHAP, and the transformer are the references interviewers themselves learned from, and recognizing them signals fluency the way no self-description can.

The map's final message is the book's own: every role in Table C.1 is being reshaped by the same forces this book has traced chapter by chapter, and the durable advantage is not any single tool but the combination the intended reader has been building all along, enough statistics to be skeptical, enough programming to be self-sufficient, and enough financial context to know what the numbers mean and what happens when they are wrong.

Wulin.Suo@Queensu.ca

Role	What the job actually is	Core chapters
AI model validator	Independently challenging machine-learning, language, and learned-policy models rather than classical ones: out-of-time and stress testing, calibration and fairness checks, evaluation of generative output, and the limits of explanation tools	6, 8, 16, 18, 19, 21
AI governance and agent supervisor	Overseeing AI agents that act rather than advise: what each agent may do, human approval points, audit trails, kill switches, and monitoring for collective behavior across agents	10, 16, 18, 19, 20, 21
LLM engineer (financial services)	Building retrieval-augmented assistants, document extraction, and agent tooling on large language models, with evaluation, guardrails, and data controls a regulator will accept	2, 6, 16, 17, 18
Payments-fraud data scientist	Scoring card and instant-payment transactions in the milliseconds before authorization, including scams in which the customer authorizes the payment, using network and behavioral features	3, 6, 15, 17, 18, 21
Climate-risk analyst	Translating physical and transition risk into credit, market, and insurance exposures: scenario analysis, carbon-price shocks, catastrophe models	9, 8, 11, 12, 17
Treasury and ALM quant	Managing a bank's or insurer's balance sheet as a whole: interest-rate risk in the banking book, deposit behavior, liquidity and funding, hedging	3, 5, 7, 8, 11
Central-bank research economist or data scientist	Forecasting, financial-stability monitoring, system-wide stress tests, and text analysis of policy communication, increasingly with agent-based and machine-learning tools	3, 7, 8, 16, 20

Table C.2: Seven roles that have taken shape as distinct jobs by 2026, and the chapters that prepare for them

Wulin.Suo@Queensu.ca